Estimation and Limit Theorems

50 minutes • 50 points • Student quiz

Name: _____ **Date:** _____

Instructions. Answer all four problems and show your reasoning. No calculator is needed; exact fractions are acceptable. Any needed normal probabilities are supplied. You may quote Chebyshev's inequality, but must derive the law of large numbers in Problem 2. Use the spaces provided; clearly label any work continued on the back. Reserve 5 minutes to check.

1. Estimating a mean and variance

10 points · about 8 minutes

A random sample of six observations from a population with finite mean μ and variance σ^2 gives the data

$$4, \quad 6, \quad 6, \quad 8, \quad 8, \quad 10.$$

(a) Estimate μ using the sample mean $\bar{x}$. Show your calculation. (3 points)

(b) Calculate the sum of squared deviations from your sample mean, $\sum_{i=1}^{6}(x_i - \bar{x})^2$. (3 points)

(c) Estimate σ^2 using the *unbiased sample variance* (divide the sum in part (b) by $n-1$). Also report the mean squared deviation (divide by n). Label the two answers. (4 points)

2. Chebyshev's law of large numbers

14 points · about 12 minutes

Let $X_1, X_2, \dots$ be independent and identically distributed, with $\mathbb{E}[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$. Put $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$.

(a) Find $\mathbb{E}[\bar{X}_n]$ and $\mathrm{Var}(\bar{X}_n)$. Show the calculation, and identify where independence is used. (4 points)

(b) Apply Chebyshev's inequality to prove that, for every fixed $\varepsilon > 0$,

$$\mathbb{P}\big(|\bar{X}_n - \mu| \geq \varepsilon\big) \longrightarrow 0 \qquad \text{as } n \longrightarrow \infty.$$

State what this says about the mode of convergence of $\bar{X}_n$. (6 points)

(c) Suppose $\sigma^2 = 4$. Find the smallest integer n for which *the Chebyshev bound* guarantees the following. (4 points)

$$\mathbb{P}\big(|\bar{X}_n - \mu| \geq \tfrac{1}{2}\big) \leq 0.05.$$

3. The central limit theorem

12 points · about 11 minutes

(a) State the usual central limit theorem for i.i.d. random variables with mean μ and variance σ^2. Give the assumption on σ^2, the standardized random variable, and its limiting distribution. A proof is not required. (4 points)

(b) Independent service times $X_1, \dots, X_{100}$ have the same exponential distribution, with mean 2 minutes and variance 4 minutes squared. Use the CLT to approximate

$$\mathbb{P}\big(1.6 \le \bar{X}_{100} \le 2.4\big).$$

You may use $\Phi(2) \approx 0.9772$ and $\Phi(-2) \approx 0.0228$, where $\Phi(z) = \mathbb{P}(Z \le z)$ for $Z \sim N(0,1)$. Show the standard error and both standardized endpoints. (5 points)

(c) Chebyshev's inequality gives

$$\mathbb{P}\big(|\bar{X}_{100} - 2| \ge 0.4\big) \le \frac{4}{100(0.4)^2} = 0.25.$$

Translate this into a lower bound for the interval probability in part (b). Explain why that bound and your CLT answer are compatible, and identify which is a guaranteed bound and which is an approximation. (3 points)

4. A heavy-tailed distribution

14 points · about 14 minutes

Let $X_1, X_2, \dots$ be i.i.d. with density

$$f(x) = \begin{cases} \dfrac{1}{2x^{3/2}}, & x \geq 1, \\ 0, & x < 1. \end{cases}$$

This is a Pareto distribution. Write $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$ and $M_n = \max\{X_1, \dots, X_n\}$.

(a) Verify that f integrates to 1, and calculate $\mathbb{P}(X_1 > t)$ for $t \geq 1$. (4 points)

(b) Evaluate the improper integral for $\mathbb{E}[X_1]$. Explain why the law proved in Problem 2 does not apply. (3 points)

(c) Fix any $A > 0$ and take n large enough that $nA \geq 1$. Use independence and part (a) to find $\mathbb{P}(M_n \leq nA)$. (3 points)

(d) Use part (c) to prove $\mathbb{P}(\bar{X}_n \leq A) \to 0$ for every fixed $A > 0$. What does this say about convergence of $\bar{X}_n$ to a finite constant? (4 points)

Hint: All observations are nonnegative, so $\bar{X}_n \geq M_n / n$. You may use $1 - u \leq e^{-u}$ for $0 \leq u \leq 1$.